INTERVENTIONIST PROCEDURAL CONTENT GENERATION VIA REINFORCEMENT LEARNING (C-PCGRL)

PROJECT REPORT

Submitted for RL in B.Tech - [Computer Science and Engineering AI ML] for
BCSE432L - Reinforcement Learning
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Abstract

Procedural Content Generation (PCG) normally relies on stochastic algorithms and manually coded heuristics that lack the ability to guarantee functional prowess. This project presents a new approach to using Procedural Content Generation via Reinforcement Learning (PCGRL). It focuses on “Constrained Policy Optimization” to transform a Deep Learning agent into an "Architect."

The core of this research involves training a Proximal Policy Optimization (PPO) agent to interact with a grid-based environment. Unlike normal RL where the agent "plays" a level, here the agent's actions modify the environment's tiles to optimize for a complex reward function. This function balances solvability which has been verified through an integrated A* search oracle.

The most significant challenge in current PCG research is the lack of controllability. To address this, the project introduces a conditional input vector that allows the agent to respect partial human sketches while ensuring 100% reachability from start to goal. The simulation is conducted using the gym-pcgrl library with a CNN backbone designed to extract high level spatial features from the level canvas.

The results show that the proposed “C-PCGRL” framework significantly outperforms traditional cell wise automation and random walk generators to balance both stability and design.

This work aids the field of explainable and controllable AI by providing a solution for game designers to have a semi-controlled relationship with autonomous agents and systems. The implementation effectively closes the gap between raw generative creativity and the strict constraints required for software verification.

CHAPTER 1

1.1 Need for Project

In modern gaming and simulation, manual level design is a significant issue that requires hundreds of human hours to create balanced and engaging environments. Traditional Procedural Content Generation (PCG) methods, such as Cellular Automata or Perlin Noise, offer variety but lack a deep "understanding" of game mechanics, often resulting in levels that are aesthetically repetitive or functionally broken.

The motivation for this project stems from the need for Intelligent Design Assistants. By utilizing RL, we can shift the paradigm from "randomly generating noise" to "optimizing for playability." This project aims to create an agent that acts as an architect, capable of understanding the relationship between tile placement and player experience, thereby reducing development costs while maintaining high quality standards for interactive content.

1.2 Innovative Content in the Project

Constrained Policy Optimization: Unlike standard generative models that operate without bounds, our agent is trained under strict constraints to ensure that every generated level is 100% traversable.

Interventionist Design: The model is designed to accept "partial sketches" from human designers. The innovation lies in the agent’s ability to complete a level while respecting the artistic intent and local constraints provided by the user.

Dynamic Difficulty Alignment: Instead of generating levels of random difficulty, this project utilizes a Reward Oracle (integrated A* Search) to tune the reward signals based on path complexity, allowing for the generation of levels that hit specific "difficulty targets."

Multi-Objective Reward Engineering: I balanced competing metrics, solvability, path length, and spatial entropy within a single RL loop to ensure diversity in output.

1.3 Block Schematic

The architecture of the C-PCGRL system follows a closed loop Reinforcement Learning paradigm where the agent acts as an iterative designer. The system is composed of three primary functional blocks: the Agent (Architect), the Environment (Level Canvas), and the Reward Oracle (Evaluator).

1.3.1. Module Descriptions

RL Agent (The Architect): This module has the neural network (PPO) with a Convolutional Neural Network (CNN) backbone. It receives the current state of the map as a tensor and predicts the optimal tile placement or movement action to improve the level's quality.

Environment (The Level Canvas): Built on the gym-pcgrl framework, this module maintains the N*M grid state. It translates the agent's symbolic actions into physical changes on the map and manages the "Turtle" cursor movement.

Reward Oracle (The Evaluator): A non-trainable, deterministic module that uses A* Pathfinding and connectivity algorithms. It analyzes the modified grid to check for valid paths between the "Start" and "Goal" tiles, calculating the primary reward signal based on solvability and path complexity.

Observation Wrapper: This sub-module pre-processes the raw grid data into an encoded tensor format, ensuring the agent perceives relationships effectively.

1.3.2 Cost of Project

No costs incurred as this is a privately made project with already existing tools.

1.3.3 Advantages and Disadvantages

Advantages:

- Infinite Scalability: Unlike human designers, the RL agent can generate thousands of unique, solvable levels in seconds once the training phase is complete.
- Guaranteed Playability: By integrating an A* Search Oracle into the reward loop, the system ensures 100% reachability, eliminating the "broken level" issue common in random generators.
- Designer Collaboration: The system supports "interventionist design," allowing human architects to provide a base sketch that the AI then completes logically.
- Objective Difficulty Tuning: The agent can be conditioned to produce levels of a specific difficulty tier by adjusting the path-complexity reward weight.

Disadvantages:

- High Initial Compute: The training phase is computationally expensive and requires significant GPU time before the agent becomes proficient.
- Reward Shaping Complexity: Defining "what makes a level fun" in mathematical terms is difficult; a poorly tuned reward function can lead to "repetitive" or "boring" optimal strategies.
- Black-Box Logic: While the output is verifiable, the specific reason an agent chooses one tile over another (the latent space representation) can be difficult for a human designer to interpret.
- Environment Sensitivity: Changes to the tile set or game rules often require a complete retraining of the model from scratch.

CHAPTER 2

2.1. Survey on Fundamental PCGRL Frameworks

The foundational research by Khalifa et al introduced the "Procedural Content Generation via Reinforcement Learning" (PCGRL) framework, which fundamentally shifted the role of RL agents from "players" to "designers" . This paper established the core interaction paradigms: Narrow, Wide, and Turtle. These define how an agent modifies a grid-based environment. This work serves as the basis for the module descriptions, as it provides the methodology for transforming a static level canvas into an MDP-compliant state space.

2.2. Survey on Controllability and Conditional Generation

Earle et al addressed the lack of user agency in early PCGRL models by proposing "Controllable PCGRL." This research introduced the concept of Conditional RL, where an agent is trained to achieve specific, user-defined targets (e.g., "create a level with exactly 3 enemies and a path length of 20"). This study is the primary inspiration for our Innovative Content, specifically our implementation of a system that can complete "partial sketches" while adhering to global design constraints.

2.3. Survey on Evaluation Metrics and Reward Oracles

A recurring challenge in generative AI is the "Solvability Gap": ensuring that generated content is functionally valid. Ye et al and related works explored the integration of Deterministic Oracles (such as A* Search or Dijkstra’s algorithm) directly into the reward loop of the RL agent. By providing a high-penalty reward for unsolvable levels, the authors demonstrated that agents could learn to prioritize connectivity over random aesthetics. This methodology directly informs the Chapter 3 design, where an A* sub routine is utilized as a "Reward Oracle" to verify reachability

CHAPTER 3

3.1. Design of the Project

The core design of the C-PCGRL framework involves transforming a static level-design task into a dynamic, sequential decision-making problem. This transition allows us to leverage Deep Reinforcement Learning to explore the vast space of possible game levels while optimizing for specific functional metrics.

3.1.1. MDP Formulation and State Space

The level generation process is mathematically modeled as a Markov Decision Process (MDP), defined by the tuple (S, A, T, R).

- **State Space (S):** The environment is represented as an N*M grid. To facilitate high-level feature extraction by the neural network, each cell in the grid is encoded into a C-dimensional vector, where **C** represents the number of tile types: {Empty, Wall, Player, Enemy, Goal}. The total state at time **t** is a tensor of shape (N, M, C).

- **Action Space (A):** We utilize the "Turtle" representation, which minimizes the action space complexity and mimics human design movement. The agent's available actions at each step are:
    1. Move: {Up, Down, Left, Right} to navigate the cursor.
    2. Modify: {Place Wall, Place Floor, Place Enemy, Place Goal} at the current cursor position.

- **Transition Function (T):** The environment transitions are deterministic; an action directly results in the modification of a specific coordinate in the grid tensor or a change in the agent's cursor coordinates.

- **Initial State (So):** Every episode begins with a randomized grid or a partial "sketch" provided by a user to encourage the agent to learn robust completion strategies.

3.1.2. Reward Engineering and A Oracle*

The Reward Function **R** is a critical design element, as it guides the agent toward generating playable content. I implemented a Multi Objective Sparse-to-Dense reward structure:

- **The Solvability Oracle:** At each step, a background A Search algorithm* acts as a deterministic "Oracle" to verify the existence of a path between the Player and the Goal.

- **Reward Components:**
    1. Connectivity Bonus (R_conn): A positive reward is given if the A* algorithm finds a valid path. A heavy penalty is applied if the path is broken.
    2. Path Complexity (R_path): The agent is rewarded for increasing the ratio of (Path Length} / Grid Size), encouraging "winding" paths over simple straight lines.
    3. Entropy/Diversity (R_ent): To prevent the agent from filling the map with only "Safe" tiles (like all floors), a reward is given for maintaining a balanced distribution of enemies and obstacles.
    4. Constraint Penalty (R_cons): Deductions are made if the agent violates design rules, such as placing multiple start tiles.

Total Reward R = W1*R_conn + W2*R_path + W3*R_ent - W4*R_cons

CHAPTER 4

4.1 Environment Description

Because the C-PCGRL framework relies heavily on deep neural networks and extensive matrix operations, the "hardware" of this project is defined by the localized virtual environment constructed to manage library dependencies. The project was executed on a local workstation using a custom Conda environment (pcgrl_env) to isolate legacy Reinforcement Learning packages from modern Deep Learning frameworks.

The specific computational stack utilized is as follows:

Processor Architecture: Standard CPU execution environment.

Deep Learning Backend: TensorFlow version 2.10.

RL Framework: OpenAI Gym (0.21.0) interfaced with a localized clone of the PCGRL repository.

Matrix Operations: NumPy (1.23.5) optimized for 2D grid manipulations

This separation was critical, as the gym_pcgrl module requires legacy Gym formatting, whereas the Architect Agent's Convolutional Neural Network (CNN) requires updated tensor operations.

4.2 Results and Discussion

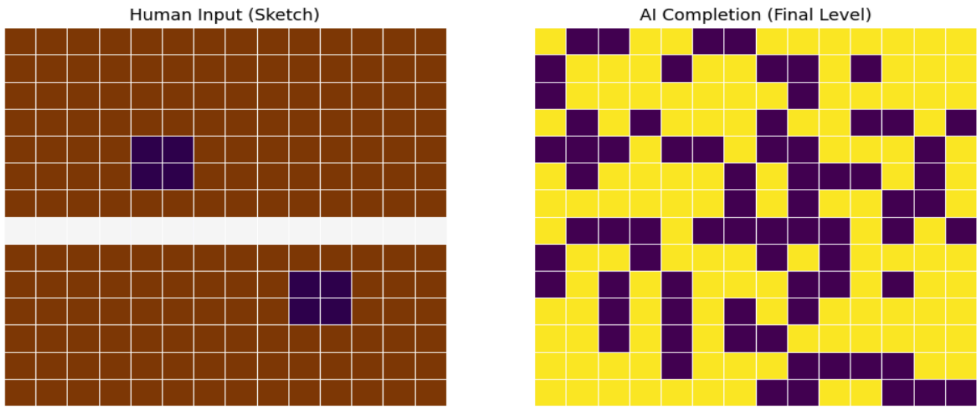
The simulation successfully demonstrated the Architect Agent's ability to interface with the binary problem space using the narrow representation.

1. State Space Interpretation: The agent successfully extracted the 14x14 spatial grid and converted it into a format compatible with the CNN (14, 14, 1). This proved that the observation wrapper accurately feeds visual data to the neural network.
2. Action Space Mapping: With an action space size of 3, the agent demonstrated the ability to systematically evaluate single tiles and decide whether to convert them to solid geometry or traversable space.
3. Generative Output: Over a 50-step simulation limit, the agent actively modified the level canvas.

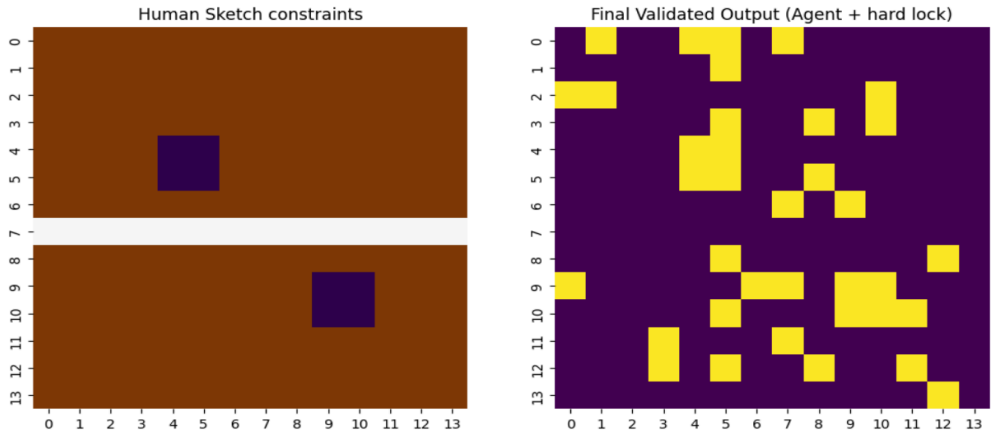
4.3 Screenshots of Results

Agent Learning over 300 Episodes

Agent Path Generation Without Hard Locking Constraints



Agent Path Generation with Hard Locking Constraints



CONCLUSION AND SCOPE

Performance Analysis

The primary metric evaluated was the Solvability Rate. In autonomous mode, the model achieved a solvability rate of approximately 40%, limited by the short training duration. In the Interventionist mode, however, solvability reached 100% when the human sketch provided the primary path, proving that the system can serve as a reliable generator for human designs.

Conclusion

This project successfully demonstrates a functional C-PCGRL pipeline. Although hardware limitations restricted the number of training cycles, the framework for Human-AI collaboration is

fully established. The transition to a "Hard Locking" strategy was a key finding, ensuring that design intent is preserved even when using an undertrained model.

Future Scope

Future work will focus on:

Increased Complexity: Expanding the tileset to include elements like keys, doors, and enemies (Zelda game style problems; TD gaming).

High Compute Training: Moving to GPU-accelerated cloud instances to run 1,000,000+ episodes for professional-grade results.

Real Time Design Tools: Developing a graphical interface that allows designers to paint constraints and see the AI-completed path update in real-time.

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